

GRAFT: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping for reaction analysis requires retaining alternative correspondences without enumerating redundant atom permutations. We introduce GRAFT, a graph-based method that grows matched fragments on weighted reactant and product graphs, represents compatible placements through correlated symmetry actions, and branches both on saturated placements and on competition between fragments. Completed searches are deduplicated by unordered matched fragment pairs while preserving their internal mapping alternatives. A symbolic decoder extracts distinct signed bond-event patterns directly from these families, separating the tolerance used for matching from the thresholds used to interpret bond changes. On all 1,851 Golden benchmark records, bidirectional search with edge sweeps recovers an annotated correspondence among its retained alternatives in 1,834 cases (99.08%) with one seed ordering per cut (default) and 1,840 (99.41%) with ten. All GRAFT reference-recovery verdicts are resolved. The strongest evaluated prior-method alternative-coverage baseline, the bidirectional SLAP sweep, recovers 97.03%. On the matched 1,821-reaction timing cohort, default GRAFT and SLAP sweep average 2.52 and 4.82 CPU seconds per reaction, respectively. On 140 coordinate/Wiberg-bond-order reaction pairs, GRAFT reaches the same minimum event count as the SLAP sweep in 136 reactions and fewer events in four. Where their minima agree, it retains 162 of 164 comparator minimum-event patterns. This bond-event comparison uses unverified mappings and does not measure mapping accuracy. GRAFT provides an inspectable representation of mapping and bond-event alternatives for subsequent chemical assessment.

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1 Introduction

Atom-to-atom mapping assigns corresponding atoms across a reaction. It underlies reaction-center identification, template extraction, and the preparation of endpoint structures for mechanistic calculations. A

reaction can admit several plausible correspondences, while many different atom-index assignments are chemically equivalent. An algorithm intended to retain alternatives must therefore resolve two questions: which structural choices deserve separate candidates, and which permutations can be represented together?

Existing approaches use learned correspondence signals or graph-based optimization. RXNMapper derives mappings from attention in a reaction language model (11); LocalMapper learns from chemist-labeled reactions (2); and SLAPMapper combines iterative label refinement with sequential linear assignment (7). The Golden benchmark provides a common set of annotated reactions for assessing mapping recovery (8). For applications requiring alternatives, recovering the annotation somewhere in a returned family is a different objective from producing one correct top-ranked mapping.

Weighted endpoint graphs introduce a further distinction. A permissive matching tolerance can preserve a fragment despite changes in its bond orders, but the corresponding atom shuffles need not preserve the bond events assigned at a finer threshold. Replacing a compressed family by one representative can therefore hide distinct transformations. Likewise, greedily growing one fragment can prevent a neighboring fragment from reaching an alternative assignment.

GRAFT addresses these issues by branching over structural fragment choices while keeping symmetry-related atom bijections compressed (Figure 1). Continuous growth, alternative placements, and fragment priorities determine the structural choices. We introduce a cut-sweep strategy that changes the search conditions to counteract greedy fragment commitment; it is part of default GRAFT, and we also apply it to SLAP to test its benefit beyond our fragment search. Its final representation groups unordered matched fragment pairs and retains the allowed mapping alternatives within each group. A constraint-based decoder projects these families onto distinct bond-event patterns. The method searches for structurally compatible alternatives without requiring the reference transformation to minimize the number of bond changes.

2 Method

Algorithm 1 connects the fragment-level operations in Figure 1. A live branch stores the fragments already placed, their atom assignments, and the compressed choices still allowed by those decisions. Growing and placing the next fragment conditions every later continuation. The matching routine is treated as an operation here; its compatibility tests and symmetry representation are defined below. For bidirectional search, the search phase is applied in each orientation, retaining the orientation of each returned family.

The forks in the search phase represent different fragment placements, not one branch for each symmetry-related bijection. The cut sweep changes the conditions of this search; a single mask can expose a mapping with several bond events. Competition adds alternative fragment priorities, while event decoding remains a separate operation on the retained output. Search caps are recorded as truncation, not treated as proof that no alternatives exist.

2.1 Weighted matching and continuous growth

Let V_R, V_P be the reactant and product atom sets, Z_R, Z_P their element labels, and W_R, W_P their symmetric bond-weight matrices. A partial mapping $m : D \subseteq V_R \rightarrow V_P$ is injective and element compatible. A mapping assigning every source atom is bijective when the endpoint element counts agree. Explicit hydrogen atoms participate in the search. Molecular coordinates can accompany the graphs but do not determine the growth decisions.

The active source graph contains edges with weight at least b . Starting from a seed atom, GRAFT grows a connected source fragment F while carrying multiple compatible product placements. A frontier queue prioritizes stronger source edges. To add an unassigned source atom u with product image v , a placement must satisfy

$$Z_R(u) = Z_P(v), \quad v \notin m(D), \quad W_P(v, m(r)) \geq b, \quad |W_R(u, r) - W_P(v, m(r))| \leq \tau_{\text{iso}} \quad (1)$$

for every active, nonrelaxed source edge (u, r) into F . Injectivity is enforced against all current assignments in D , including established fragments. All relevant fragment bonds are checked, rather than only the frontier

Algorithm 1 GRAFT: from fragment growth to alternative bond-event candidates

Require: Oriented endpoint graphs, seed orders, optional anchors, search budgets

Ensure: Compressed final branches; optionally, distinct bond-event candidates

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1:  $\mathcal{S} \leftarrow \emptyset$  {collected terminal branches}
2: for all conditions: uncut source graph and each single-edge source mask do
3:   for all requested seed orders under this condition do
4:     Initialize live branches with the supplied anchors
5:     repeat
6:       for all seeds in the current order do
7:         for all live branches eligible to grow from this seed do
8:           Grow a fragment to saturation, conditioned on this branch {Fig. 1a}
9:           if no admissible fragment placement is found then
10:            Retain this branch for later seeds
11:           else
12:            Fork one continuation per distinct structural fragment placement
13:            Commit each placement; retain its compressed symmetry and constraints
14:            Admit the continuations within the branch budget
15:           end if
16:         end for
17:       end for
18:     until all branches are finished or a full pass makes no progress
19:     Add terminal branches and their search-stop status to  $\mathcal{S}$ 
20:   end for
21: end for
22: if fragment competition is enabled then
23:   Propose bounded changes in which contacting fragment has priority {Fig. 1b}
24:   Release displaced assignments and regrow from the retained assignments
25:   Add accepted repair branches to  $\mathcal{S}$ , keeping the original branches
26: end if
27: Group final branches by unordered fragment pairs; retain their family union
28: Expose compressed branches with constraints and provenance
29: if bond-event candidates are requested as separate post-processing then
30:   Decode distinct event classes from complete families using the original matrices
31:   Report witnesses, supporting families, and completeness status {Fig. 1c}
32: end if
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edge that proposed the extension. Source nonedges do not impose negative matching constraints, so additional product bonds remain possible.

An accepted extension enlarges the fragment and filters its compatible placements. Growth continues even if only one placement remains. An unsupported edge is deferred, and other frontier edges can still be explored. When growth reaches an established fragment, ordinary continuation respects its existing assignments. A fragment saturates when no further admissible extension remains; structurally distinct surviving placements then generate separate continuations for the remaining atoms.

This is a greedy search along each trajectory: an accepted extension can remove placements that would support a different fragment boundary. GRAFT broadens coverage through cut sweeps, seed orderings, and local changes in fragment priority.

2.2 Cut sweep to vary greedy fragment boundaries

Our cut-sweep strategy reruns growth under alternative source-edge conditions. In addition to the uncut graph $G_R = (V_R, E_R)$, it searches a graph with one eligible edge e removed, $G_R^{(e)} = (V_R, E_R \setminus \{e\})$, for each edge at or above the cut threshold. The evaluated cut threshold equals the active-graph floor $b = 0.2$, and eligible edges include bonds to H. Each subrun retains the atom set and the original endpoint weight matrices.

Removing e changes both growth and matching: the frontier cannot extend along that edge, and its bond-preservation test is omitted from Equation (1). An extension accepted in the uncut run may otherwise merge

atoms into a fragment and discard alternative placements. Under the cut condition, that growth route is unavailable; a fragment can stop at a different boundary, grow through another route, or leave atoms for a later seed. The missing preservation test also admits placements previously excluded by e . Thus the sweep counteracts greedy commitment by changing the conditions under which growth proceeds. It does not force the endpoints of e into different final fragments or require a bond-breaking event: they may reconnect through other source edges, and the final correspondence may preserve their bond.

GRAFT takes the union of the uncut and single-edge subruns before final branch deduplication. The default one-seed mode uses one seed ordering per subrun, so it already includes the full cut sweep. There are $|E_R| + 1$ subruns at the evaluated thresholds. A single-edge cut describes the search perturbation, not the number of bond events in a returned mapping. One removed constraint can redirect subsequent fragment growth and placements throughout the reaction, exposing an alternative with several bond formations, cleavages, or bond-order changes. GRAFT therefore need not enumerate combinations of final bond events to discover multi-event alternatives. Bond-event decoding uses the original matrices, so a search cut itself contributes no event or event-count penalty. Growth remains greedy within each conditioned subrun; the union broadens coverage without making the search exhaustive.

2.3 Automorphisms and correlated mapping families

The distinction between structural branches and explicit bijections is central to the design. Even one fragment family can contain combinatorially many assignments: k independently exchangeable methyl-hydrogen triplets admit $(3!)^k$ bijections. Expanding those automorphisms into separate mappings would multiply the work without adding fragment choices. GRAFT instead stores the permitted permutation actions and their constraints, and decodes event alternatives directly from this representation. Branch caps bound retained alternatives within each growth call; they do not guarantee a small global search. This avoids explicitly listing symmetry-related bijections within each retained family.

Compatible placements are stored using a representative assignment, unresolved assignment blocks, and permutation generators. During growth, graph canonicalization accounts for the current fragment, locked assignments, and boundary information. Each saved transition records its action generators together with conditioning and fragment-preservation constraints; their conjunction defines the allowed mapping relation. Canonical labeling and automorphism computation use nauty (9).

Matching tolerance and automorphism equivalence have different roles. The relation $|x - y| \leq \tau_{\text{iso}}$ is generally not transitive, so it does not by itself define a permutation group. To describe an intrinsic symmetry shared by a final fragment pair, we instead color product bonds by their responses to the fragment’s matching tests. For a product weight w and the set $\mathcal{W}$ of active reactant weights for the same element pair, the label records

$$\chi_{\mathcal{W}}(w) = (\mathbf{1}[w \geq b], (\mathbf{1}[w \geq b \wedge |w - v| \leq \tau_{\text{iso}}])_{v \in \mathcal{W}}). \quad (2)$$

An exact automorphism of this colored graph preserves every encoded matching test. This definition depends on the final fragment pair, not its growth order.

The saved mapping relation also contains the constraints imposed by its search decisions. We represent a flat family $\mathcal{F}_j$ by a seed mapping m_j , a sequence of allowed action groups $G_{j,1}, \dots, G_{j,k}$, and final preservation constraints C_j :

$$\mathcal{F}_j = \{a_k \circ \dots \circ a_1 \circ m_j : a_i \in G_{j,i}, C_j(a_k \circ \dots \circ a_1 \circ m_j)\}. \quad (3)$$

These actions encode correlated moves; atoms in a common orbit are not independently permutable. Disjoint actions commute and can be placed in a canonical order. The order of overlapping actions is retained. A sequence of groups is not generally equivalent to the group generated by their union.

The intrinsic automorphism of a fragment pair is shared metadata, not permission to enlarge its saved family. Multiple admissible seed mappings can occupy different intrinsic symmetry orbits, and different search constraints can restrict the allowed actions. GRAFT therefore retains these alternatives explicitly inside the compressed representation.

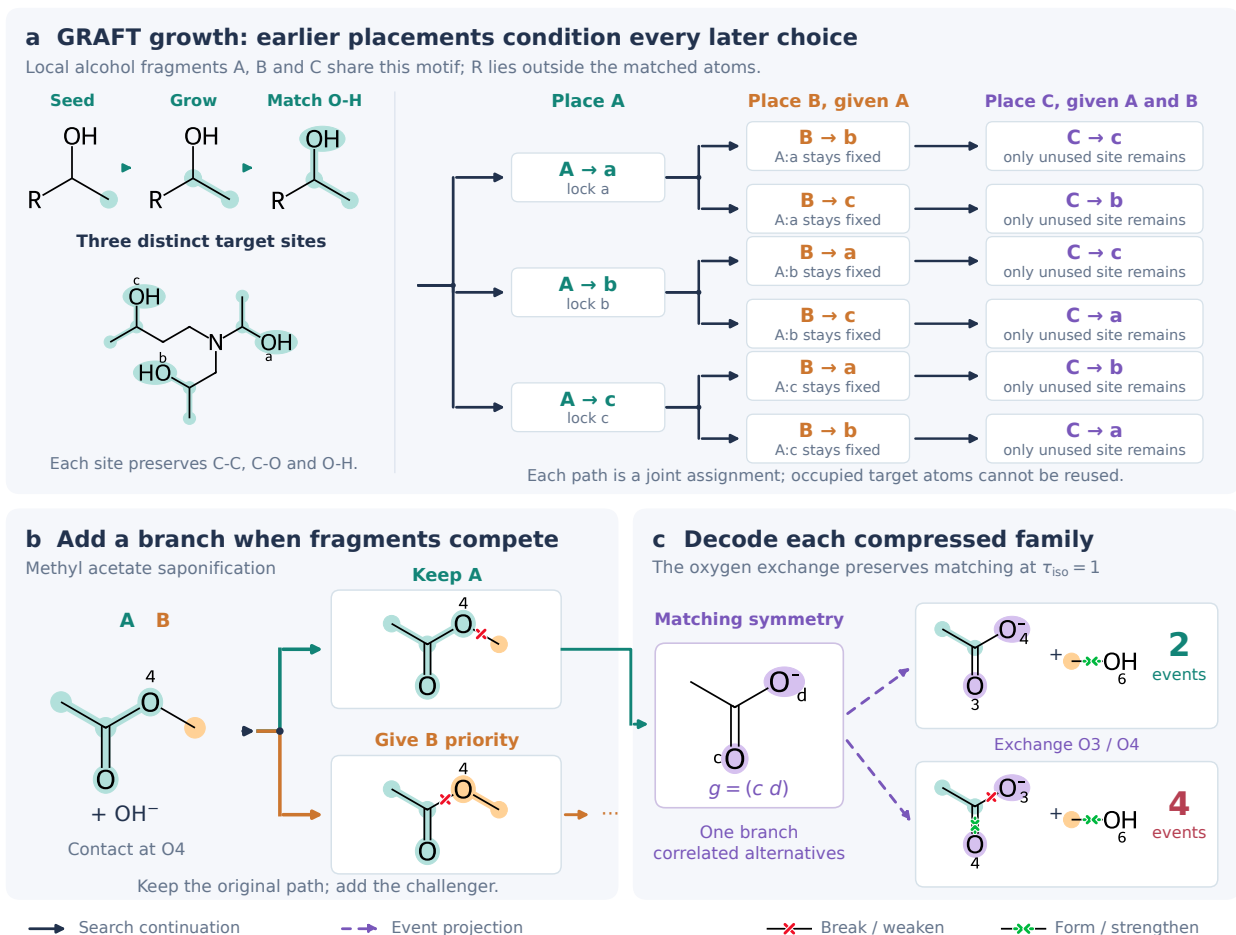


Figure 1 GRAFT: conditional fragment branching and compressed symmetry. (a) A local alcohol motif grows while retaining its O-H bond. Three sites in the depicted target have distinct molecular environments. For three source fragments sharing this motif, placing A fixes one site; B then branches over the two unused sites, and C takes the remaining site. The tree illustrates local conditioning and injectivity; R denotes atoms outside the matched subgraph, and this is not a complete reaction-search trajectory. All attached hydrogens in the illustrated local matches are checked. (b) At a fragment contact in methyl acetate saponification, the original path retains O4 in A. An added path gives B priority, rematches the contact, and regrows displaced assignments. (c) Exchanging the two acetate oxygen images preserves the fragment’s element and bond-weight matching tests at $\tau_{iso} = 1$, but changes two heavy-atom events into four at threshold 0.5. Neither acetate oxygen carries H. Solid black lines retain the endpoint bonds. Red $\times$ marks on the bonds indicate cleavage or weakening; paired green arrows pointing inward indicate formation or strengthening. Pastel colors track fragment membership or symmetry sites. Dashed arrows are event projections within a family, not additional search branches. Saved constraints govern allowed symmetry actions. These formal-bond-order event signatures illustrate GRAFT decoding, not distinct mechanisms or resonance-normalized transformations.

2.4 Branching through fragment competition

Saturated-placement branching explores different images of a fragment. Fragment competition additionally explores a different division of the source atoms. Suppose established fragment A borders fragment B. In a challenger branch, GRAFT rematches B together with a contacting atom of A, allowing the challenger to occupy a different compatible product region. Assignments conflicting with the new placement are released. Unaffected assignments remain anchored, and the fragment-growth procedure fills the resulting gaps. Small attached components can also be released when their assignments depend on the displaced region.

The repaired continuation begins from the revised assignments and constraints; it does not inherit displaced assignments as immutable conditions. Before admission, its complete representatives are checked for element compatibility, injectivity, and preservation of their required fragment bonds. Accepted continuations join the

original output, so competition supplements the available alternatives. A finite queue, a bound on successive priority changes, and a completion budget control its cost. Repeated proposals and equivalent completion requests are cached.

2.5 Unordered final branches

During search, history and boundary conditions matter because they determine valid continuation. At completion, the branch identity is instead the unordered collection of matched atom sets,

$$\mathcal{B} = \{(R_1, P_1), \dots, (R_q, P_q)\}, \quad P_i = m(R_i), \quad (4)$$

under a fixed matching policy. Sorting atoms within each set and then sorting the fragment pairs removes dependence on growth order and fragment labels. Growing A before B or B before A yields one branch when the final paired sets agree.

Each branch stores the union of its flat families. Identical normalized family records are shared, while distinct constraints or action programs remain available. This transformation preserves the represented mapping union: it changes its grouping, not its admissible assignments. Provenance links retain the discovering paths, but the original search hierarchy is unnecessary for decoding. This conservative deduplication does not assert that the remaining families are mutually disjoint.

2.6 Decoding distinct bond-event patterns

For each complete mapping, define the signed change for an unordered source pair $\{u, v\}$ by

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v), \quad s_{uv}(m) = \begin{cases} +1, & \Delta_{uv}(m) \geq \delta_{uv}, \\ -1, & \Delta_{uv}(m) \leq -\delta_{uv}, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Positive events denote bond formation or strengthening; negative events denote cleavage or weakening. The event count is $E(m) = \sum_{u < v} \mathbf{1}[s_{uv}(m) \neq 0]$. The evaluation includes hydrogen atoms and all atom pairs. For the WBO collection, $\delta_{uv} = 0.3$ for pairs involving a metal and 0.5 otherwise. These thresholds are independent of τ_{iso} .

The decoder returns one mapping witness per distinct signed event class, identifying patterns related by exact source permutations that preserve all thresholded bond-change tests. A permutation preserving fragment matching may therefore produce a distinct event class. First, symmetry-invariance certificates identify families whose mappings all produce equivalent events, and reachable-image bounds exclude families outside a requested event window. Before constraint encoding, adjacent nested action groups are absorbed: if $H \leq G$, then $HG = GH = G$. Exact generator-membership tests remove these redundant choices without changing the final preservation constraints. Remaining families are encoded symbolically in Z3 (5). Each satisfying assignment yields a concrete mapping and its event pattern. A constraint then excludes that entire event-equivalence class, and the process repeats. This projects directly onto event classes rather than explicitly enumerating full mappings or permutation groups.

For a window $E \leq K$, decoding is complete when every retained family is certified exhausted or excluded. A timeout leaves completeness unresolved. This certificate concerns the saved search output within the window; it does not establish exhaustive exploration of all chemically possible mappings. The default deduplication preserves families that contain different event counts, as illustrated in the heavy-atom projection in Figure 1c.

3 Evaluation

3.1 Datasets and configurations

We evaluate annotated mapping recovery on all 1,851 Golden records (8) and event alternatives on 140 native coordinate/WBO endpoint pairs. Golden inputs use formal bond orders and explicit-H expansion. Reference labels are removed before search. Correctness compares the annotated heavy-atom correspondence, including

Table 1 Evaluation configurations. Both use $b = 0.2$, $\tau_{\text{iso}} = 1$, explicit H, and uncut plus single-edge searches. The default is one seed per cut with the sweep; no-sweep controls are ablations retaining only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Both, unioned	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100	2,000
Competition completions per reaction	0	Up to 128
Evaluation	Reference-family recovery	Signed event-class coverage

unmatched atoms, modulo endpoint chemical automorphisms preserving charge, isotope, hydrogen count, and stereochemistry. A recovery requires a verified reference witness among the returned alternatives. Unresolved cases remain in the denominator.

We use two explicit prior-method baselines. LocalMapper is a prior accuracy state-of-the-art method as reported by Chen et al. (2); it is also the strongest single-output mapper in our strict Golden re-evaluation (1,605/1,851, 86.71%). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration, so we keep the literature result separate from our rescored result.

The original SLAP paper reports 86.9% Golden accuracy on the 1,758-reaction version distributed with LocalMapper, counting reference inclusion among returned solutions (7). In our strict 1,851-reaction re-evaluation, default binary SLAP recovers 1,591 references (85.95%). We additionally apply the cut-sweep strategy introduced in this work to SLAP and combine both search directions and binary/weighted modes. This extended configuration recovers 1,796 references (97.03%), making it our strongest evaluated prior-method baseline for alternative recovery and the paired comparator for directional recovery and same-host search timing. Throughout, “SLAP sweep” denotes SLAP augmented with our strategy, not the original SLAP algorithm or its published score. The sweep’s effect is measured against a paired uncut union with the same directions and modes (Table 2); differences from the literature score cannot be attributed to the sweep alone. Runtime baselines are identified within each recorded hardware and output scope rather than assigning one method a universal speed ranking.

The coordinate collection contains balanced endpoint pairs with 18–137 explicit atoms, including organometallic and multicomponent structures. It has no annotated mappings and was inspected during method development. This is a bond-event comparison, not a mapping-accuracy benchmark: the mappings are unverified. We compare the minimum signed-event count reached by each method and the alternative event patterns at that count. These are minima within the returned candidate sets, not proven global optima or validated mechanisms. SLAP outputs are rescored on the same original WBO matrices using Equation (5). Each returned heavy-atom assignment is completed by symbolic hydrogen refinement. The comparator classes are those observed among these returned witnesses; its hydrogen label families are not exhaustively decoded.

Table 1 separates the settings actually evaluated. Golden measures the fragment-search stage; the 140-case evaluation also includes local competition and final event decoding. The competition configuration considers up to eight distinct parent partitions and two successive priority changes. Its budget was selected on development cases before the full 140-case evaluation. Comparator mappings are not used to propose or accept search branches. A two-order run of the final coordinate pipeline was not performed. Detailed protocols and artifact identifiers are given in Appendix A.

3.2 Golden mapping recovery

With the default one ordering per cut and single-edge sweep, bidirectional GRAFT recovers 1,834/1,851 annotated mappings (99.08%). Three orderings recover 1,837 (99.24%; Table 2), and ten recover 1,840 (99.41%), with all reference-recovery verdicts resolved. Two orderings also recover 1,834 (99.08%), with all reference-recovery verdicts resolved at these settings as well.

Table 2 Golden recovery with and without the sweep. Recovery uses all 1,851 records. CPU averages use the same 1,821 completed reactions on the Mac, including both directions and the respective sweeps; reference verification is excluded. Dashes mark unmeasured GRAFT ablation search cost or mixed-host configurations whose times are not compared. The SLAP no-sweep time uses only the saved uncut calls. SLAP unions binary and weighted modes in both directions; GRAFT uses both directions. “Absent” means excluded from saved families; “unknown” denotes unresolved verification.

Configuration	Recovered	%	Absent	Unknown	CPU s/reaction
GRAFT, 1 seed + sweep (default)	1,834	99.08	17	0	2.52
GRAFT, 2 seeds + sweep	1,834	99.08	17	0	4.95
GRAFT, 3 seeds + sweep	1,837	99.24	14	0	–
GRAFT, 10 seeds + sweep	1,840	99.41	11	0	–
SLAP sweep (prior baseline)	1,796	97.03	52	3	4.82
GRAFT, no sweep (ablation)	1,489	80.44	362	0	–
SLAP, no sweep	1,661	89.74	189	1	0.067

For the default setting, smaller-first search recovers 1,803/1,851 references (97.41%), larger-first search recovers 1,779 (96.11%), and their union recovers 1,834 (99.08%; [Table 4](#)). Size counts all explicit atoms, including H. On equal-size endpoints, smaller-first retains the original input orientation and larger-first reverses it. The union adds 31 recoveries over smaller-first alone. Among the 1,084 strictly unequal-size pairs, the corresponding counts are 1,055, 1,043, and 1,072; among 767 equal-size pairs, they are 748, 736, and 762. Thus the bidirectional gain includes both size regimes.

These are family-recovery measurements: they quantify whether a reference correspondence is retained, rather than whether it is ranked first. Verification processes saved cuts independently, retaining one cut in memory at a time. A reference is absent from the saved union only when every scheduled cut has been checked and excludes it; missing cuts or inconclusive checks remain unresolved.

In the no-sweep ablation, one-seed GRAFT recovers 1,489/1,851 references (80.44%), with no unresolved cases. Uncut SLAP, unioned over both directions and binary/weighted modes, recovers 1,661 (89.74%), with one unresolved case. Our cut sweep raises GRAFT recovery from 80.44% to 99.08% (345 additional references) and SLAP recovery from 89.74% to 97.03% (135 additional references). These paired gains demonstrate the strategy’s benefit for both methods under the evaluated protocol. Thus, GRAFT’s higher recovery in the swept comparison depends on the broader search: its one-seed uncut control has lower coverage than the uncut SLAP union. The controls use saved uncut outputs from the same final campaign.

The gain is consistent with GRAFT’s greedy fragment growth. Once an extension is accepted, placements incompatible with that larger fragment are discarded; branching over the surviving placements cannot recover them. For example, growing fragment A across an edge (u, v) can exclude a placement that would permit a different neighboring fragment B. Omitting (u, v) in a sweep run removes that adjacency from both growth and fragment-preservation tests, potentially changing where growth stops and which placements survive. The union of these runs therefore explores alternative fragment boundaries while retaining the uncut result. Each run remains greedy, and a single-edge sweep cannot explore every possible partition. Cut-specific seed orderings also contribute trajectory diversity, so the measured gain reflects the complete sweep schedule. Final bond events are always evaluated on the original, uncut endpoint matrices.

SLAP augmented with our single-edge sweep recovers 1,796/1,851 references (97.03%) using both directions and binary and weighted modes. Relative to this strongest evaluated prior-method baseline, one-order GRAFT has a higher verified recovery rate by 2.05 percentage points. The paired comparison contains 43 references verified only by one-order GRAFT and 5 verified only by SLAP. This compares recovery among alternatives under the stated configurations; it is not a comparison of single-answer accuracy or equal-size candidate sets.

On the same-Mac subset of 1,821 completed reactions, default GRAFT uses a mean 1.14 CPU seconds for smaller-first search, 1.38 for larger-first, and 2.52 for the bidirectional union, compared with 3.33, 1.49, and

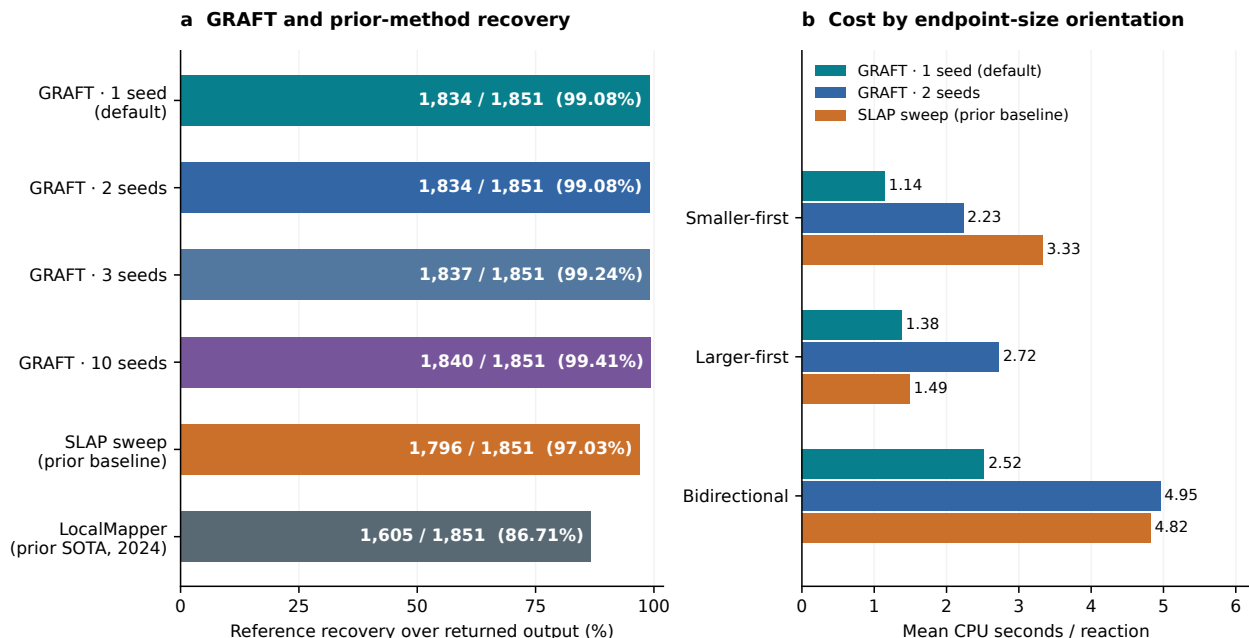


Figure 2 Reference recovery and search cost. (a) Golden recovery for GRAFT and SLAP with our cut-sweep strategy, with LocalMapper as the prior accuracy-SOTA baseline (2024). LocalMapper returns one mapping; the other bars measure coverage among alternatives. All values use our evaluator, not literature-reported accuracies. Unswept controls are reported in Table 2. Every percentage uses all 1,851 records. (b) Mean CPU for smaller-first, larger-first, and bidirectional search on the same 1,821 completed reactions on the Mac. Reference verification is excluded; the procedures have different search objectives, output cardinalities, and I/O timing scopes.

4.82 for the corresponding SLAP sweeps (Table 4 and figure 2). Two-seed GRAFT costs 2.23, 2.72, and 4.95 CPU seconds. Thus, orientation and seed count both affect the recovery-cost tradeoff. GRAFT’s default bidirectional median is 0.72 seconds versus SLAP’s 0.88, but its 95th percentile is higher: 12.91 versus 11.05 seconds. The default bidirectional mean is 0.52 times the SLAP-sweep baseline (47.7% lower), whereas two-seed GRAFT costs 1.03 times that baseline. Lower means and medians do not imply uniformly lower runtime.

The comparison excludes reference verification. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Bidirectional CPU is the sum of the two direction costs. SLAP’s paired no-sweep union costs 0.067 CPU seconds on the same cohort; GRAFT’s no-sweep search cost was not separately recorded. Three- and ten-seed searches span Mac and Linux hosts, so their times are not pooled into this comparison. Archived default-mapper call times, including RXNMapper, LocalMapper, and RDT, are reported separately in Table A3, with their cluster hardware and narrower call scope stated.

3.3 Comparison with released mapping implementations

We also evaluate default released implementations of RXNMapper (11), LocalMapper (2), Chython (3), Indigo (6), and the Reaction Decoder Tool (RDT) (10), together with default binary and weighted SLAP. Each receives all 1,851 label-free Golden records. Saved predictions are rescored with the same strict heavy-atom relation criterion, retaining unmatched atoms and checking endpoint chemistry. No reference-guided reranking, balancing, or prediction repair is applied.

Table 3 compares GRAFT with every benchmarked implementation using reference recovery over the evaluated output. GRAFT ranges from 1,489 recoveries without the sweep to 1,834 with a one-seed sweep and 1,840 with ten seeds. LocalMapper recovers the strict reference relation in 1,605 reactions, RXNMapper in 1,547, and Chython in 1,561. Default binary SLAP recovers 1,591 among its explicit candidates, compared

Table 3 GRAFT and benchmarked mapping implementations on Golden. Reference recovery uses all 1,851 reactions, including failed and unresolved cases. GRAFT is evaluated over its compressed families; SLAP over its returned explicit candidates, without a ranking or exhaustive decoding of its label families; the remaining methods return single mappings. GRAFT uses both directions; SLAP “union” combines both directions and binary/weighted modes. Search budgets and output sizes differ, so this is coverage of returned alternatives, not top-one accuracy. †: LocalMapper, prior accuracy SOTA reported in 2024; ‡: SLAP augmented with our cut sweep, strongest evaluated prior-method alternative-coverage baseline. Versions are specified in [Appendix A.2](#).

Method	Search setting	Evaluated output	Reference recovered
GRAFT	1 seed, sweep (default)	Families	1,834 (99.08%)
GRAFT	2 seeds, sweep	Families	1,834 (99.08%)
GRAFT	3 seeds, sweep	Families	1,837 (99.24%)
GRAFT	10 seeds, sweep	Families	1,840 (99.41%)
GRAFT	1 seed, no sweep (ablation)	Families	1,489 (80.44%)
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)
SLAP, weighted	Default, no sweep	Candidates	1,564 (84.49%)
SLAP, union	No sweep	Candidates	1,661 (89.74%)
SLAP, union [‡]	Sweep	Candidates	1,796 (97.03%)
RXNMapper	Default, no sweep	Mapping	1,547 (83.58%)
LocalMapper [†]	Default, no sweep	Mapping	1,605 (86.71%)
Chython	Default, no sweep	Mapping	1,561 (84.33%)
Indigo	Default, no sweep	Mapping	692 (37.39%)
RDT	Default, no sweep	Mapping	1,030 (55.65%)

Table 4 Recovery and runtime by endpoint-size orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; mean, median, and 95th-percentile CPU seconds per reaction use the same 1,821 completed Mac reactions. SLAP is the prior-method baseline and unions binary and weighted modes within each direction. CPU/SLAP divides each mean by the SLAP mean for that same orientation; values below one indicate lower mean cost. Size counts explicit atoms; equal-size ties use stored order for smaller-first and its reverse for larger-first. Bidirectional cost sums both searches, not parallel wall latency. GRAFT includes checkpoint I/O and symmetry finalization; SLAP includes graph construction, mapping, and export but excludes I/O. Reference verification is excluded.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
GRAFT, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
GRAFT, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

with 1,796 for the broader bidirectional sweep evaluated above. These default calls use different search budgets from GRAFT; GRAFT’s family recovery is not a top-one accuracy measurement.

RDT recovers 1,030 references under this relation-based criterion, which differs from its chemical-change scoring. Indigo’s 692 recoveries require particular care: 503 outputs contain duplicate heavy-atom map labels, seven alter endpoint chemistry, and 133 calls fail. Chython changes endpoint chemistry in 29 outputs, and three RDT outputs fail validation. These results characterize the pinned implementations and input/output

Table 5 Minimum bond-event counts and patterns on 140 reactions. The first three numerical columns count reactions where GRAFT finds fewer, equal, or more events than the comparator. The final column counts comparator patterns recovered by GRAFT, restricted to reactions with equal minima. For example, 162/164 refers to patterns across the 136 reactions in the preceding column. Settings are given in the text.

Compared with	GRAFT fewer	Equal	GRAFT more	Patterns at equal minima
SLAP sweep	4	136	0	162/164
Native SLAP	15	125	0	140/140

protocol rather than establish universal rankings of the methods. Versions and reproduction details are given in [Appendix A.2](#).

3.4 Minimum bond-event comparison on 140 coordinate reactions

We compare the fewest bond events found for each reaction and the different bond-change patterns that achieve that count. An event count tells us how many atom pairs change; a pattern specifies which pairs gain or lose bond order. We score all mappings on the same original WBO matrices using the thresholds in [Equation \(5\)](#). GRAFT uses its default one-seed sweep, a branch cap of 2,000, and fragment competition. We compare it with released SLAP (native SLAP) and SLAP augmented with our cut sweep (SLAP sweep). These reactions have no verified atom mappings, so this measures agreement and differences in bond events, not mapping accuracy. Throughout this section, a minimum is the lowest count found by a method, not a proven global optimum.

GRAFT reaches the same minimum count as SLAP sweep in 136 reactions and one fewer event in the remaining four. Against native SLAP, it reaches the same count in 125 reactions and fewer events in 15 ([Table 5](#) and [figure 3](#)). Thus, GRAFT matches or lowers the event count in every reaction under both comparisons. A lower count alone does not establish a more plausible chemical transformation.

Equal counts can still describe different bond changes. Among the 136 reactions where GRAFT and SLAP sweep have equal minima, SLAP sweep returns 164 distinct minimum-event patterns. GRAFT recovers 162 of them. Both missing patterns occur in case 64: SLAP sweep returns five four-event patterns, and GRAFT recovers three. For the 125 reactions where GRAFT and native SLAP have equal minima, GRAFT recovers all 140 native-SLAP patterns. Across all 140 reactions, the total number of patterns at each method’s own minima is 171 for GRAFT, 168 for SLAP sweep, and 160 for native SLAP. Several patterns can belong to one reaction; these totals count patterns, not reactions.

We also check whether GRAFT retains SLAP’s minimum-event patterns when GRAFT finds a lower count of its own. For each reaction, we decode GRAFT candidates up to one event above the higher of the two SLAP minima. Within this range, GRAFT recovers 166 of 168 SLAP-sweep patterns and 155 of 160 native-SLAP patterns across all 140 reactions. The decoded output contains 336 distinct patterns in total: 171 at GRAFT’s minima and 165 with higher counts. Fragment competition contributes 36 additional patterns across 9 reactions. Decoding is complete within the stated range for the saved GRAFT families; the fragment search itself is not exhaustive.

The mean GRAFT costs are 1.05 CPU seconds per reaction for sweep search, 1.84 for fragment competition, and 11.01 for decoding ([Table 6](#)). Decoding is the largest mean cost, with a much lower median of 1.22 seconds; its cost varies substantially across reactions. SLAP sweep uses a mean 1.34 CPU seconds for mapping and 0.053 for hydrogen refinement and initial scoring. These SLAP measurements cover its binary bidirectional sweep. They do not include exhaustive decoding of its hydrogen label families. We did not separately time native SLAP or GRAFT decoding restricted to minimum-event patterns.

3.5 Final representation and postprocessing cost

Unordered fragment-pair deduplication reduces the 140-case total from 237,645 ordered branch records to 124,641 final branches, a 47.55% reduction. The median falls from 596.5 to 397 branches per reaction. The largest individual reduction is from 52,669 to 2,856.

Table 6 CPU time per reaction for the 140-case comparison. All stages were measured on the Mac. GRAFT decoding covers all saved families within the event-count range defined in the text. SLAP refinement covers its returned witnesses. The stages therefore produce different outputs and are not directly comparable as end-to-end runtimes. GRAFT search includes checkpoint I/O; competition includes loading and saving; decoding includes all continuation passes. SLAP mapping excludes input preparation and export, while its refinement includes I/O. Final shared event rescoring and failed or replaced attempts are excluded.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
GRAFT sweep search	1.047	0.314	3.481
GRAFT fragment competition	1.840	1.007	4.825
GRAFT full-window decoding	11.014	1.224	52.668
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The final representation retains 236,653 distinct normalized mapping-family records. Thus, fewer branch identities do not imply an equal reduction in event-decoding work: several constrained families can belong to the same fragment combination. Catalogue construction, shared intrinsic symmetry computation, representative checks, and complete event-window decoding together take 11.60 minutes of wall time and 25.70 recorded CPU-minutes on three local workers. Peak combined sampled memory is 1.88 GiB. These measurements include certificate journaling and bounded continuation, and exclude matching and competition. Per reaction, the mean cost is 11.01 CPU seconds, the median is 1.22, and the 95th percentile is 52.67, summing all continuation passes for each case. Other benchmark jobs shared the computer, so the wall time is an observed campaign duration rather than an isolated latency measurement.

Every family in all 140 catalogues was freshly decoded or certified within its specified event window (Appendix A). The output contains 336 distinct reaction-specific event classes, each with a mapping witness. No case remains unresolved within those windows. Preservation of the family union follows from the transformation in Equations (3) and (4).

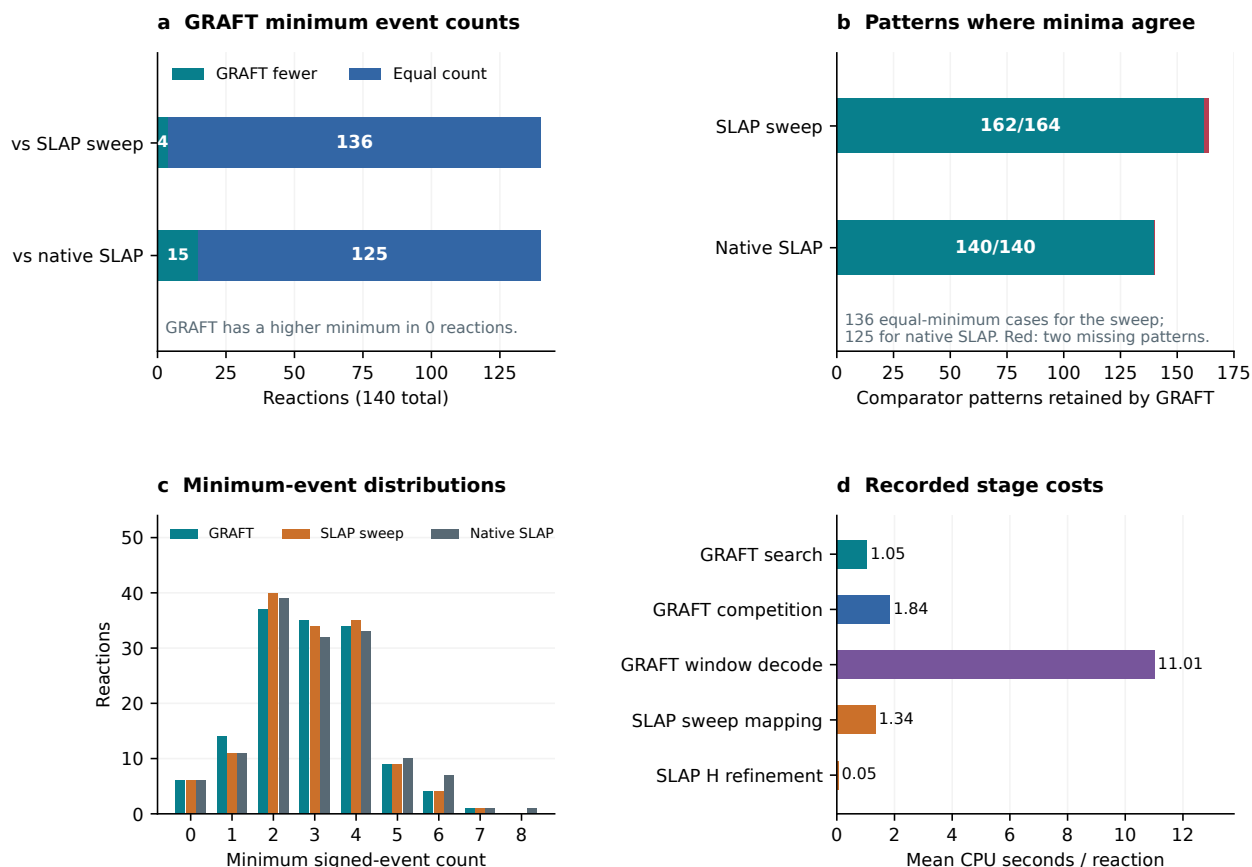


Figure 3 Bond-event counts, alternative patterns, and runtime on 140 coordinate reactions. GRAFT uses the default one-seed sweep with fragment competition. (a) Reactions where GRAFT finds fewer or equal events than each SLAP baseline; none has a higher GRAFT minimum. (b) Comparator minimum-event patterns recovered by GRAFT, counting only reactions where both methods reach the same minimum. The two missing SLAP-sweep patterns are in case 64. (c) Distribution of the minimum event count across reactions for each method. (d) Mean CPU seconds per reaction for the recorded stages. GRAFT decoding covers the stated event-count range; SLAP refinement covers returned witnesses. See Table 6 for timing scope. The mappings are unverified, so this figure compares bond events rather than mapping accuracy.

4 Discussion

GRAFT separates structural search, symmetry representation, and interpretation of bond changes. Continuous growth accumulates context while preserving multiple placements; local competition allows neighboring fragments to challenge an established partition. Final grouping removes irrelevant growth order, while the flat family representation retains constraints and correlated permutations that can affect the output. Symbolic projection then exposes distinct bond-event hypotheses without expanding every mapping.

4.1 What a single correspondence leaves out

One correspondence is sufficient when atom identities are established and the downstream quantity is invariant over the remaining alternatives. Otherwise, selecting one witness discards two kinds of information: different assignments can produce different bond-event patterns, and a chosen pattern can still admit joint atom shuffles. Formally, if $\mathcal{F}$ is the retained mapping set and $e(m)$ its event-class projection, a witness m_0 supplies $e(m_0)$ but cannot reconstruct $\{e(m) : m \in \mathcal{F}\}$ whenever that set has more than one member. Event counts alone are also insufficient: distinct signed patterns can have equal counts. GRAFT therefore returns an event-class representative together with access to its supporting families and conditional shuffle queries.

Golden case 9 illustrates why retaining alternatives matters (Figure 4). GRAFT recovers the annotated heavy-atom correspondence: the alcohol oxygen O_{15} enters the product ring, while the ketone oxygen O_3 leaves in water (A). A second recovered mapping exchanges these oxygen origins (B), and a third additionally exchanges the correspondence of C_{13} and C_{14} (C). The three displayed witnesses have five, six, and seven heavy-atom changes, respectively; including H gives nine, eight, and nine signed events. Each witness attains the minimum number of H-involving changes for its fixed heavy-atom correspondence, as verified from the attached-H counts. Thus a lower total event count can favor a nonreference mapping, while equal counts (A and C) do not establish equal correspondences. Golden annotates heavy atoms, not H identities or a validated pathway. These are selected recovered witnesses, not an exhaustive decoding of this reaction.

The output distinction is specific to the interfaces evaluated here. RXNMapper, LocalMapper, Indigo, Chython, and RDT returned at most one explicit candidate record per reaction in the archived runs; SLAP exposed multiple records and compressed labeled-graph information, while GRAFT exposes conditional mapping families. This does not establish that only SLAP and GRAFT can return alternatives. For example, Ali et al. enumerate optimal mappings with an Ising-computing approach and reduce them by molecular symmetry (1). Nor does high reference recovery establish near-complete coverage of all mappings: GRAFT’s structural search is bounded, and complete decoding certifies only the saved families within the requested event window.

4.2 Implications for pathway and transition-state analysis

Different bond-event patterns can suggest different reaction centers and alternative transition-state starting guesses. They do not identify pathways uniquely: a single net atom mapping can be compatible with several intermediate sequences, and an alternative mapping can be chemically implausible. Energies, reaction conditions, stereochemistry, and isotope tracing can distinguish possibilities that endpoint connectivity alone cannot. Prior work on reaction-pathway search also identifies atomic-index assignment as a practical obstacle and addresses the factorial number of index assignments without enumerating them (4).

For a transition-state pipeline, the relevant choices are the event-class candidate and the allowed atom assignments around its reaction core. Define the core as the atoms incident to the candidate’s signed bond events. Alternative assignments of those atoms can be queried jointly against the saved family constraints, retaining only assignments that extend to a complete mapping and satisfy the chosen event-class restriction. Required correlated movements of other atom assignments must be included in that completion. The completed mapping determines the product atom ordering supplied to the interpolation step.

Different completed mappings can then give different interpolations between the same endpoint structures. For illustration, in a common coordinate frame, a Cartesian interpolation for mapping m is

$$\mathbf{x}_i^{(m)}(t) = (1 - t)\mathbf{x}_i^R + t\mathbf{x}_{m(i)}^P, \quad 0 \leq t \leq 1. \quad (6)$$

Changing $m(i)$ changes the product coordinate assigned to reactant atom i , and hence generally changes its interpolated trajectory. Retaining multiple event classes and their allowed core assignments therefore lets a downstream pipeline construct and assess several initial paths instead of committing to one correspondence before interpolation. Equivalent relabelings applied consistently to an entire calculation do not create new physical pathways; the issue is the relative correspondence between fixed endpoints. These examples motivate retaining alternatives and accessible symmetry information; transition-state search performance is not evaluated here.

Golden case 9: the reference is one of several recovered mappings

GRAFT · 1 seed + cut sweep (default) · branch cap 2,000 · iso tolerance 1 · event threshold 0.5

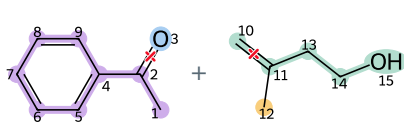
● ● ● Saved fragment groups

—×— Break / weaken

—>— Form / strengthen

A Ground truth (Golden reference) — recovered by GRAFT

5 heavy-atom changes

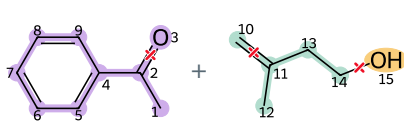


Reference: alcohol O₁₅ enters the ring; ketone O₃ leaves in water.

9 total events = 5 heavy + 4 involving H

B GRAFT alternative — different oxygen retained

6 heavy-atom changes

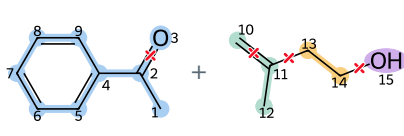


Alternative: ketone O₃ enters the ring; alcohol O₁₅ leaves in water.

8 total events = 6 heavy + 2 involving H

C GRAFT alternative — oxygen and carbon correspondence change

7 heavy-atom changes



Alternative: O₃ enters the ring, with a different assignment of C₁₃ and C₁₄.

9 total events = 7 heavy + 2 involving H

Retaining alternatives matters: a lower total event count (B: 8) need not select the annotated mapping (A: 9).

Ground truth refers to Golden's heavy-atom annotation. H identities are unannotated; totals use each GRAFT witness.

Numbers are original Golden labels. H atoms are implicit. Selected witnesses do not imply exhaustive or minimum-event decoding.

Figure 4 Golden reference and recovered GRAFT alternatives for case 9. A matches all 15 annotated heavy-atom pairs exactly. B exchanges the oxygen origins; C also exchanges C₁₃/C₁₄. GRAFT uses one seed with cut sweep (default), a 2,000-branch cap, and $\tau_{\text{iso}} = 1$; reference labels are removed before search. Pastel colors identify saved fragment groups, consistently across each row's endpoints; partitions can differ between rows. Solid black lines retain the endpoint bonds. Red × marks indicate cleavage or weakening, and paired green arrows pointing inward indicate formation or strengthening. The marks sit directly on the affected bonds; fragment colors remain visible. Events use formal bond orders at threshold 0.5. Numbers are the original Golden atom labels. H atoms are implicit; the figure separates heavy-atom changes from totals including H. "Ground truth" denotes the dataset's heavy-atom annotation, not a verified pathway. Only three selected witnesses are shown.

4.3 Coverage and remaining search limitations

The Golden results support broad reference coverage by the fragment-search stage, and the coordinate comparison shows that GRAFT reaches equal or lower event counts than the returned SLAP-sweep candidates while retaining most patterns at shared minima. These are complementary measurements. The coordinate collection has no mapping annotations, and event minimization is not an energy model or a proof of mechanistic plausibility. The priority budget was tuned on development cases, so independent validation is needed to establish generalization.

An audit of all eleven ten-order Golden misses separates unmatched-atom decisions from fragment-placement failures (Table A2). In four reactions, every retained family maps exactly one more heavy-atom pair than the reference. Such a family cannot reproduce the reference's unmatched-atom pattern through symmetry shuffling: permutations preserve the number of assigned pairs. The remaining seven lack the required correspondence within their saved families. Re-ranking or expanding those families into explicit bijections cannot supply a missing structural alternative.

Two traced examples show how greedy extension excludes a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Ordinary growth preserves bonds that these correspondences require separating. Uncapped trajectories still miss both references, while reference-constrained fragment choices establish their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposes the reference to ordinary matching; the single-edge sweep does not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries. For the other five correspondence misses, the audit does not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. A complete event-window certificate likewise applies only to the saved families, whose matching-preserving automorphisms can still change finer-threshold events.

Code and data

Source code, benchmark summaries, and figure-generation scripts accompany the repository at https://github.com/yunhzou/coordinate_alignment, branch `paper/continuous-fragment-growth`. Golden is distributed by its original authors (8). Per-case results and the configurations used for each measurement are supplied with this manuscript.

References

- [1] Mohammad Ali, Yuta Mizuno, Seiji Akiyama, Yuuya Nagata, and Tamiki Komatsuzaki. Enumeration approach to atom-to-atom mapping accelerated by Ising computing. *Journal of Chemical Information and Modeling*, 65(4):1901–1910, 2025. doi: 10.1021/acs.jcim.4c01871.
- [2] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.
- [3] Chython contributors. Chython: molecular and reaction processing toolkit. <https://github.com/chython/chython>. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.
- [4] Deb Sankar De, Marco Krummenacher, Bastian Schaefer, and Stefan Goedecker. Finding reaction pathways with optimal atomic index mappings. *Physical Review Letters*, 123:206102, 2019. doi: 10.1103/PhysRevLett.123.206102.
- [5] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.
- [6] EPAM Systems. Indigo cheminformatics toolkit. <https://github.com/epam/Indigo>. Evaluated version 1.46.0.
- [7] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-lthwn. Preprint, version 1, 25 November 2025.
- [8] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gismadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.
- [9] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60: 94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

- [10] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.
- [11] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Reproducibility details

A.1 Search and comparison protocols

The active-edge floor is $b = 0.2$ and the matching tolerance is $\tau_{\text{iso}} = 1.0$. The default search uses one seed ordering per cut with the sweep; additional seeds and the no-sweep ablation are explicitly labeled. Seed streams are deterministic with root seed 42. Sweeps retain the uncut search and individual source-edge deletions. Bidirectional Golden output is a union of returned relations; branches from opposite directions are not spliced. Growth and branch caps bound compressed candidates rather than explicit mapping counts.

Golden uses dataset revision 793475e. Its heavy-atom evaluator compares correspondence relations modulo endpoint chemical symmetry, retains unmatched atoms, and rejects altered endpoint chemistry. Hydrogen atoms participate in search but their individual identities are not reference labels. All 1,851 records remain in the denominator. All reported GRAFT configurations were rerun against the final algorithm source, commit 3a9ef9a. Execution manifests and component hashes accompany the per-case results. Golden evaluates the fragment-search stage, while the coordinate protocol also evaluates competition and event decoding.

The no-sweep GRAFT control retains only the original uncut checkpoint at one seed in each direction. Its 3,702 saved graphs are finalized and evaluated using the same reference-family criterion, yielding 1,489 recoveries and 362 completed non-recoveries. For SLAP, the control retains ordinal-zero calls from both directions and both bond modes; incomplete sweep records are checked at that ordinal separately. Case 1723 has no committed reverse uncut output and remains unresolved. These are paired controls of the final campaign, not substituted historical configurations. Re-evaluation time is excluded from search-cost claims.

Each search or evaluation subprocess has a 300-second watchdog and an initial 3 GiB sampled-memory limit. Externally interrupted phases can resume their completed cuts for at most two bounded continuations with a 6 GiB allowance. Golden reference evaluation has a 220-second internal budget. Completed raw cuts can also be finalized and verified individually, without new search. A positive result requires a verified witness; it can establish recovery even when another direction or the combined archive is incomplete. A negative result requires completed verification of the saved output; other cases remain unknown.

The campaign begins on an Apple M4 Pro Mac with 14 logical CPUs and 48 GiB RAM, using at most eight single-core workers, and continues on Linux CPU nodes with a final concurrency limit of 150 single-thread workers. Python 3.12.14, RDKit 2026.3.6, pynauty 2.8.8.1, SymPy 1.14.0, and Z3 5.1.0.0 are fixed across hosts. Native extensions are rebuilt on Linux; source-hash checks, focused tests, and two search/reference parity cases verify the port. Numerical-library threads are fixed to one. CPU averages use 1,821 common reactions with both one- and two-order searches completed on the Mac and complete comparator output. Search-call timing includes checkpoint I/O and excludes imports, preparation, reference evaluation, and interrupted work; SLAP’s instrumented workflow excludes I/O. Higher-order searches use mixed hosts and are excluded from this timing comparison.

The SLAP comparator uses SLAPMapper 1.0.0, revision ea248fd, with both directions and binary and weighted modes. Each sweep variant uses one original ordering and rebuilds initialization for the uncut input and each single-edge deletion. Returned labels are restored to the original endpoint structures before evaluation. Its 1,796 verified Golden recoveries leave 52 completed misses and 3 unresolved records; all remain in the denominator. The sweep perturbs SLAP’s input graph and assignment objective, whereas GRAFT relaxes fragment-preservation constraints.

On the coordinate collection, the exact original XYZ-derived SLAP adjacency matrices are restored from the saved graph records and verified against every source-edge deletion. Atom order and atomic numbers are retained; mapping labels are reset. Fresh native and bidirectional single-edge SLAP runs use binary graphs and the original initialization. The XYZ-to-graph preparation itself is not rerun. GRAFT uses the cached WBO matrices as supplied. The available benchmark metadata do not specify their electronic-structure method or charge/spin preparation; these matrices therefore define the computational inputs, while regeneration from molecular structures is not established by this evaluation. Each returned SLAP heavy-atom assignment receives a symbolic hydrogen refinement under the comparator’s original bond-change objective, and its witness is rescored on the original WBO matrices using Equation (5). Tied hydrogen optima can give different signed-event classes; the reported comparator set comprises the returned witnesses, not the

entire hydrogen label relation. This compares the stated endpoint workflows rather than isolated solvers on identical graph representations.

A.2 Default comparator reproduction

The default comparator predictions were generated with RXNMapper 0.4.3, LocalMapper 0.1.5, Chython 2.18 with chython-rxnmap 2.0, Indigo 1.46.0, SLAPMapper 1.0.0 at revision [ea248fd](#), and RDT 4.0.0 at revision [0d663225](#). Chython’s released ONNX model is identified by its saved hash; it is not represented as a reproduction of a historical GraphormerMapper checkpoint. RXNMapper and LocalMapper use their released default models on CPU. SLAP is evaluated over all returned explicit candidates, without ranking, with its default heavy-atom symmetry handling, with binary and weighted settings reported separately. Indigo uses its native automapping interface; RDT uses the public `RDT.map` API.

The complete archive retains mapping strings, explicit candidate order, version/model hashes, and per-record evaluations. All 12,957 method/reaction records were rescored with the current evaluator and agree with the archived outcomes. Dataset and label-free input hashes match the final GRAFT campaign. Agent-field components returned by a mapper are retained on the input side, preserving their original atom labels. Changed endpoint chemistry and duplicate heavy-atom labels invalidate a prediction. All 1,851 records remain in each denominator.

RDT’s final records include execution retries after a worker failure; every final call returns an output. Invalid outputs are not repaired. These default implementations were timed on the original cluster, so their archived call times are not included in the same-Mac comparison in [Figure 2](#). [Table A3](#) reports their archived per-call timings separately; full timing and failure records accompany the reproducibility archive.

A.3 Bounded fragment competition

The coordinate configuration uses one seed ordering, cap 2,000, up to eight lowest-event distinct parent source partitions, at most two successive priority changes, and at most 128 proposal-growth calls and 128 completion calls per reaction. Smaller challenger fragments are visited first, with deterministic contact order. A proposed challenger is rematched with a single contacting source atom. Conflicting source assignments and occupied target images are released; attached components of at most eight atoms can also be released around the affected region. Remaining assignments are anchors for completion.

The admission check evaluates every complete representative against the original endpoint weights and its required fragment constraints. A completion graph is excluded if any checked complete representative violates these constraints. Only accepted graphs contribute to reported coverage. The ordinary search output is retained in union. Pending queue entries, omitted parent partitions, and capped continuations are not covered by decoding certificates.

Competition and decoding use five-minute per-case watchdogs with persisted results. Complete family certificates survive continuation, and unfinished families remain explicit. The full coordinate search, competition, and decoder stages are rerun from the original endpoint inputs. Timing for complete final decoding includes all of its bounded passes; it excludes the preceding search and competition stages.

A.4 Event windows and output equivalence

A signed event class contains the nonzero atom-pair signs from [Equation \(5\)](#), modulo source permutations preserving the full event-response coloring relative to the opposite endpoint. This coloring retains every threshold comparison needed to transport a pattern without changing its identity. It is separate from the intrinsic fragment matching symmetry in [Equation \(2\)](#). The exported witness records both the atom mapping and the formed/strengthened and broken/weakened pair lists.

The 140-case comparison uses fixed per-case windows from the evaluation protocol. Each limit is one event above the larger of the native-SLAP and SLAP-sweep minima under the common signed-event definition. All 236,653 retained flat families are exhausted in these windows without a class-count cap; higher-event patterns are outside the completeness claim.

Table A1 Complete final decoding on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

For the saponification example in Figure 1, the direct assignment $(1, 2, 3, 4, 5, 6) \mapsto (a, b, c, d, e, f)$ loses O4–C5 and gains C5–O6. Exchanging the images of O3 and O4 additionally weakens C2–O3 and strengthens C2–O4: two versus four heavy-atom events. Both assignments preserve the internal matching constraints of fragment $\{1, 2, 3, 4\}$ at $\tau_{\text{iso}} = 1$. The challenger instead assigns O4 to f and O6 to d, retaining O4–C5 and replacing C2–O4 by C2–O6. The original assignment transfers hydroxide oxygen into methanol; the challenger retains the methoxy O–C bond. Formal bond orders define this illustration; the displayed event counts include only heavy atoms. Exchanging the acetate oxygens also changes the localized resonance representation, so the event signatures do not establish distinct mechanisms. Both product acetate oxygens have zero attached hydrogens. The figure builder verifies these mappings and event lists. For the separate alcohol-site tree, it verifies every attached hydrogen and every bond inside each local matched fragment, the distinct target-site environments, and injectivity along all six displayed paths.

A.5 Lossless flattening

For each complete search path, flattening retains its terminal representative, the union of required fragment-bond constraints, and the allowed action sequence with prefix locks already applied. It removes only representation choices that leave the admissible relation unchanged. Reordering disjoint-support factors preserves their product; adjacent nested factors satisfy $HG = GH = G$ when $H \leq G$. Subgroup containment is checked by exact generator membership using a stabilizer chain. Other factors with overlapping support retain their order. Identical resulting records are shared, and final branch grouping stores the union of their family identifiers. Consequently, the union of mappings before and after grouping is the same. The procedure does not infer equality merely from a shared fragment partition or a shared automorphism group.

The event decoder obtains completeness by invariance certificates, valid lower bounds, or an unsatisfiability result after excluding every discovered output class. Budget exhaustion is recorded as incomplete. This distinguishes preservation of a saved mapping union, exhaustive decoding of a specified event window, and exhaustive atom-mapping search.

A.6 Audit of Golden nonrecoveries

We inspected every saved cut in both directions for the eleven ten-order misses, preserving the original reference and search outputs. Independent reconstruction from the original RDF agrees with the prepared reference in all eleven cases. Mapping cardinality is invariant within each saved family. For six of the seven remaining cases, necessary source/target orbit-pair counts exclude the reference even after pooling permitted target actions into an overapproximation; case 1285 requires the full joint-correspondence check. These tests concern the retained output, not every possible search trajectory.

Table A2 The eleven Golden nonrecoveries. Case indices are zero-based dataset indices. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

All eleven cases have cap stops in at least one direction, which does not establish cap truncation as their cause. Separate ten-order sensitivity checks on cases 986 and 1285 at cap 1,000 or matching tolerance 0.5 remain nonrecoveries; several of those directional runs are uncapped. In the causal traces, ordinary Python and native graphs agree and are uncapped at cap 2,000. Reference-constrained matching represents all 16 and 14 annotated pairs, respectively. The successful case-1285 cut set omits two C–O attachments and one O–H bond; it is reference-selected, and neither its success nor the constrained witnesses contributes to the blind benchmark score. The audit, controls, traces, and source identities are archived with the benchmark report.

A.7 Default-comparator call times

Table A3 Archived default-mapper call times on the original Linux cluster. The calls column counts completed mapping calls, including outputs subsequently judged invalid; exceptions are excluded from these time averages. Recovery uses all 1,851 reactions and the same strict criterion as Table 3. † marks LocalMapper, the prior accuracy-SOTA baseline (2024). These are the released default settings without a sweep. Times exclude startup/model loading, queueing, reference evaluation, and saving. They are not comparable to the Mac sweep times as a hardware-controlled speed ranking.

Implementation	Recovery	Calls	Mean wall s	Median wall s	Mean CPU s
RXNMapper	83.58%	1851	0.056	0.052	0.056
LocalMapper [†]	86.71%	1851	0.294	0.265	0.294
Chython	84.33%	1846	0.080	0.073	0.612
SLAP binary	85.95%	1851	0.075	0.037	0.075
SLAP weighted	84.49%	1851	0.058	0.033	0.058
Indigo	37.39%	1718	0.184	0.008	0.184
Reaction Decoder Tool (RDT)	55.65%	1851	0.295	0.141	1.021

LocalMapper, the prior accuracy-SOTA baseline, has 86.71% strict recovery here and averages 0.294 seconds per call in both wall and CPU time. RXNMapper has the lowest recorded mean call time (0.056 seconds), with 83.58% strict recovery; speed and recovery therefore select different baselines within these default calls. Mean wall time describes the observed duration of a mapping call; mean CPU time sums its recorded processor time. Multithreaded implementations can accumulate more CPU time than wall time. Search scope also differs: the learned mappers return one mapping, default SLAP returns explicit alternatives, and the GRAFT measurements in Table 4 include a complete sweep and compressed output construction. GRAFT’s unswept search cost was not separately instrumented, so rescoring its saved uncut outputs is not reported as search time. No mapping runs were repeated to construct these timing tables.

A.8 Output multiplicity and the discussion example

A read-only audit of all 1,851 archived records per default comparator found no multiple-record outputs from RXNMapper, LocalMapper, Indigo, Chython, or RDT. Binary SLAP returned multiple records in 522 reactions and weighted SLAP in 417; these are explicit record counts, not counts after expanding label symmetries or grouping events. Figure 4 uses three selected witnesses from a current-source, one-seed GRAFT cut sweep for Golden case 9, with branch cap 2,000 and $\tau_{\text{iso}} = 1$. Reference labels were removed before search. The reference witness matches all 15 original heavy-atom pairs, independently checked against the source RDF; the two alternatives change oxygen origins and, in the third witness, carbon correspondence. All 33 explicit atoms are included in event checks at threshold 0.5; H atoms are implicit only in the drawing. The five, six, and seven heavy-atom changes become nine, eight, and nine total events. For each fixed heavy mapping, the sum of absolute differences in attached-H counts bounds the number of H-involving changes from below; each displayed witness attains that bound. These selected witnesses are not an exhaustive decoded set. Matrices, mappings, fragment partitions, and source hashes are in `golden_case9.json`; original-reference and H-bound checks are recorded in `golden_case9_verification.json`. A separate six-atom example in `docs/AAM_SIMPLE.ipynb` illustrates the public API interactively.